

Capstone Project: - Insurance Price Prediction

Business Problem

We all know that Health care is a very important domain in the market. It is directly linked with the life of the individual; hence we have to always be proactive in this particular domain. Money plays a major role in this domain, because sometimes treatment becomes super costly and if any individual is not covered under the insurance, then it will become a pretty tough financial situation for that individual. The companies in medical insurance also want to reduce their risk by optimizing the insurance cost, because we all know a healthy body is in the hand of the individual only. If individuals eat healthy and do proper exercise the chance of getting ill is drastically reduced.

Problem Statement

The objective of this exercise is to build a model, using data that provide the optimum insurance cost for an individual. You have to use the health and habit related parameters for the estimated cost of insurance

File: Data.csv & **Target variable:** insurance_costv

Data Dictionary

- **Applicant_id** - Applicant unique ID
- **Years_of_insurance_with_us** - Since how many years customer is taking a policy from the same company only
- **Regular_checkup_lasy_year** - Number of times customers have done the regular health checkup in the last one year
- **Adventure sports** - The customer is involved with adventure sports like climbing, diving, etc.
- **Occupation** - Occupation of the customer
- **Visited_doctor_last_1_year** - Number of times a customer has visited doctor in the last one year
- **Cholesterol_level** - Cholesterol level of the customers while applying for insurance
- **Daily_avg_steps** - Average daily steps walked by customers
- **Age** - The age of the customer
- **Heart_decs_history** - Any past heart diseases
- **Other_major_decs_history** - Any past major diseases apart from the heart like any operation
- **Gender** - The gender of the customer
- **Avg_glucose_level** - Average glucose level of the customer while applying for the insurance
- **Bmi** - BMI of the customer while applying for the insurance
- **Smoking status** - Smoking status of the customer
- **Year_last_admitted** - When customer was admitted in the hospital last time
- **Location** - Location of the hospital
- **Weight** - Weight of the customer

- **Covered_by_any_other_company** - Customer is covered by any other insurance company
- **Alcohol** - Alcohol consumption status of the customer
- **Exercise** - Regular exercise status of the customer
- **Weight_change_in_last_one_year** - How much variation has been seen in the weight of the customer in the last year
- **Fat percentage** - Fat percentage of the customer while applying for the insurance
- **Insurance cost** - Total Insurance cost

Instructions: -

The capstone will consist of the following deliverables,

1. Milestone 1 (40 marks)

- Data Report
 - Import the given database and store it in a data frame.
 - Print shape and size of dataset
 - Understand and print attributes info (variable info, renaming if required)
- Initial Exploratory Data Analysis
 - Perform Uni-variate analysis on all features and write your observation for every analysis. (distribution and spread for every continuous attribute, distribution of data in categories for categorical ones)
 - Perform Bi-variate and multivariate analysis on different attributes then record your findings for each analysis. (relationship between variables, correlations)
- Data pre-processing
 - Check for the duplicates and non-relevant columns and drop if there are any.
 - Check for unwanted characters in the dataset and treat them appropriately.
 - Check and print the number of missing values if there are any and treat them (if required).
 - Visualize and print the outliers then treat the outliers (if required).
 - Perform Variable transformations and new variable creation is required
- Exploratory Data Analysis
 - Use different insightful visualizations and EDA techniques and report relationships among variables, and important variables
 - Perform normalization or scaling if required.

2. Milestone 2 (40 marks)

- Model Building
 - Split the dataset into dependent and independent variables.
 - Build one base model on the training dataset.
 - Evaluate the model's performance using appropriate metrics on both train and test datasets.
 - Build other (advanced) models and evaluate the model on both train and test datasets.
 - Apply all the possible tuning techniques to train the best model for the given data. Select the final best-trained model with your comments for selecting this model and deploy the model using frameworks like streamlit.
 - Write conclusions and Interpretations from the best-performing model.
- Business Insights and Recommendations
 - Mention your Actionable insights for the final selected model.
 - Write recommendations for the business

Final Presentation (20 marks)

- Executive Summary
- Data Preprocessing and Exploratory Data Analysis
- Model Building and Evaluation
- Showcase deployment on streamlit